

Case 02: Total Hip Replacement System Design Using SolidWorks

1. Introduction

Total Hip Replacement (THR) is an orthopedic surgical procedure in which a damaged hip joint is replaced with artificial prosthetic components. The procedure is commonly performed in patients suffering from osteoarthritis, rheumatoid arthritis, avascular necrosis, hip fractures, and severe joint degeneration. The primary objective of THR is to restore mobility, reduce pain, and improve the patient's quality of life.

2. Objective

- To develop a 3D CAD model of a Total Hip Replacement System using SolidWorks.
- To understand the geometry and assembly of hip implant components.
- To create an implant assembly suitable for visualization and future manufacturing studies.
- To study the design features of orthopedic hip prostheses.

3. Software Used

Component	Software
CAD Modeling	SolidWorks
Assembly Design	SolidWorks
Visualization	SolidWorks

4. Components of Total Hip Replacement System

Component	Function
Acetabular Cup	Replaces the natural hip socket
Acetabular Liner	Provides low-friction articulation
Femoral Head	Acts as the ball of the joint
Femoral Stem	Transfers load to the femur

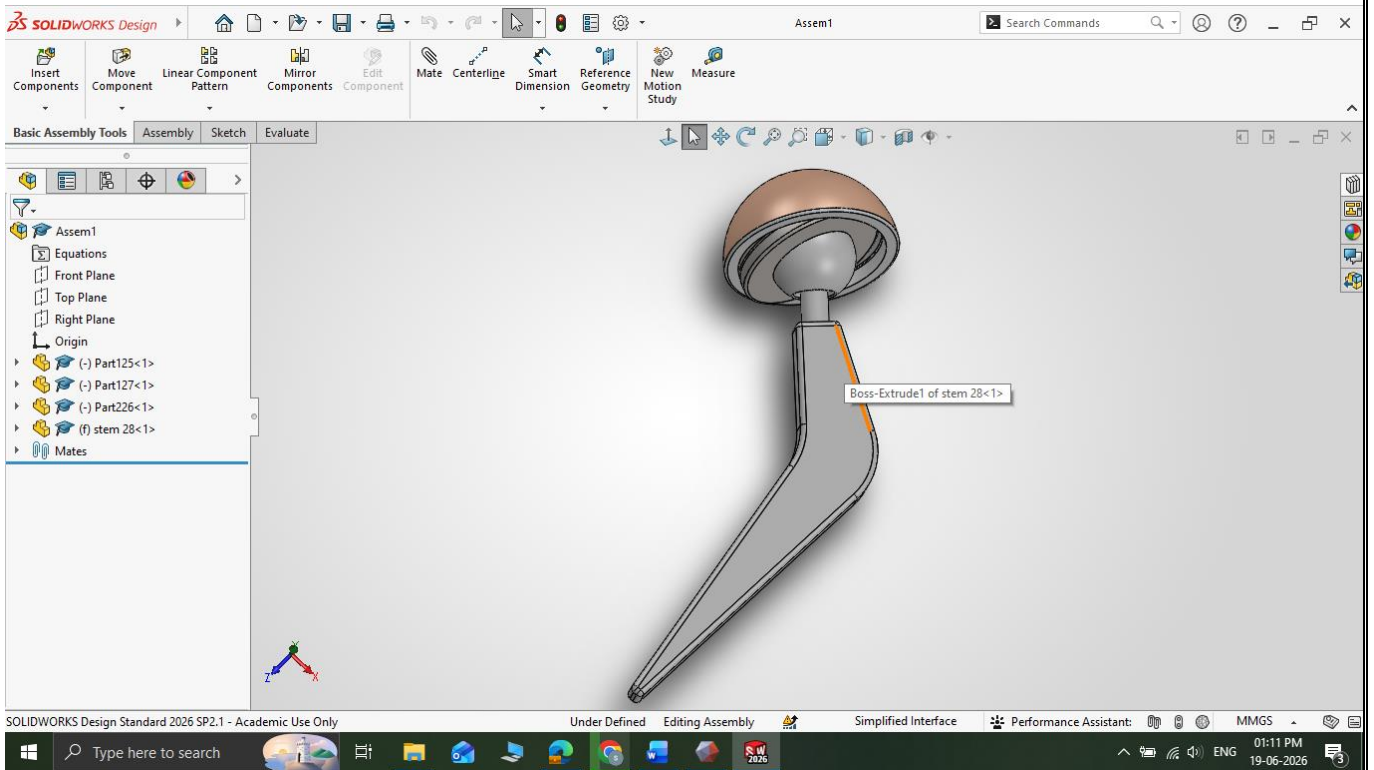
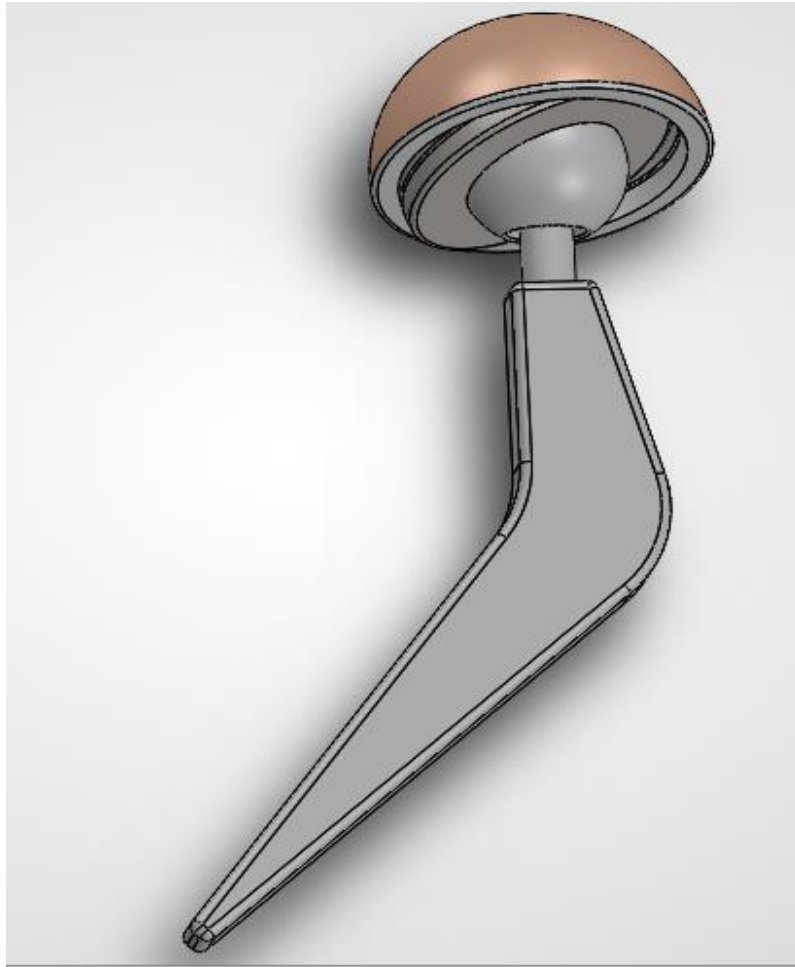


Table: Materials Used in Total Hip Replacement System

Component	Material	Reason for Selection
Femoral Stem	Ti-6Al-4V Titanium Alloy	High strength, corrosion resistance, excellent biocompatibility
Femoral Head	Cobalt-Chromium-Molybdenum (CoCrMo) Alloy	High wear resistance and hardness
Acetabular Shell (Cup)	Ti-6Al-4V Titanium Alloy	Lightweight, strong, promotes osseointegration
Acetabular Liner	UHMWPE (Ultra-High Molecular Weight Polyethylene)	Low friction and excellent wear properties
Bone Screws (if used)	Titanium Alloy (Ti-6Al-4V)	Biocompatibility and corrosion resistance
Porous Coating (Optional)	Titanium Porous Layer / Hydroxyapatite Coating	Enhances bone ingrowth and implant fixation